Duc Doan Sinh

EMBEDDED SYSTEMS & IOT STUDENT | FOCUSED ON ARTIFICIAL INTELLIGENCE

Hanoi, Vietnam

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Education

Hanoi University of Science and Technology (HUST) *Hanoi, Vietnam*

B.S. IN SMART EMBEDDED SYSTEMS AND IOT (EXPECTED 2027) *Sep. 2023 - Present*

- CPA: 3.16/4.0

The University of Tokyo, Matsuo-Iwasawa Laboratory *Tokyo, Japan*

EXCHANGE STUDENT - SHORT-TERM AI RESEARCH EXCHANGE *Mar. 2025*

- Explored advanced AI research directions, focusing on World Models, Large Language Models and Robotics.
- Delivered a final presentation and incorporated expert feedback from Matsuo-Iwasawa Lab members.

Research Experience

EDABK Laboratory, HUST *Hanoi, Vietnam*

LAB MEMBER *Sep. 2024 - PRESENT*

- Research, implement emerging AI architectures (SNN, KAN) and optimization techniques (MLP NAS) for biological signal processing (ECG, PPG).
- Develop and deploy Computer Vision systems using deep learning models, with a focus on efficiency and accuracy.
- Mentor junior lab members in machine learning fundamentals and support their onboarding into the laboratory.

Work Experience

Matsuo-Iwasawa Laboratory, The University of Tokyo *Remote*

TEACHING ASSISTANT (PART-TIME) *Apr. 2026 - PRESENT*

- Facilitate online Office Hours for the GCI course, addressing student queries on machine learning concepts and programming assignments.
- Organize and moderate virtual networking sessions to foster student engagement and build collaborative study communities across diverse international groups.

Viettel Telecom, Viettel Military Industry and Telecoms Group *Hanoi, Vietnam*

ARTIFICIAL INTELLIGENCE INTERN (PART-TIME) *Apr. 2026 - PRESENT*

- Design an LLM-powered support assistant using Retrieval-Augmented Generation, including document chunking, retrieval pipeline design, prompt iteration, and response-quality evaluation for in-app query resolution.

HANET Technology, G-Group *Hanoi, Vietnam*

ARTIFICIAL INTELLIGENCE INTERN (PART-TIME) *Apr. 2026 - PRESENT*

- Optimize YOLO-based object detection inference pipelines for deployment, profiling latency and throughput while improving model efficiency under real-time constraints.

Honors & Awards

2026	**2nd Place Winner (Vietnam Hub) & Top 100 Global Teams** HSIL Hackathon 2026 – Building High-Value Health Systems: Leveraging AI *Selected from 14,700+ applications worldwide*	*Harvard Health Systems Innovation Lab*
2025	**Outstanding Student** Global Consumer Intelligence Course 2025 *Selected as one of the Top 20 most outstanding students*	*Matsuo-Iwasawa Lab, UTokyo*
2025	**Top 30 Team** Bosch | CodeRace Challenge 2025	*Bosch Global Software Technologies Vietnam*

Projects

Neuromorphic Ear-to-Chest ECG Reconstruction with Perceptual Loss *AI Researcher / Project Lead*

DOAN-DUC/ECG_PERCEPTUAL *May 2026 - Present*

- Developed a quantized Spiking Neural Network autoencoder for reconstructing clinical-style chest ECG from noisy ear-worn ECG signals using 4-bit LSQ Conv1D layers, MultiSpike activations, and fixed-point input quantization.
- Designed a medical time-series training objective combining MSE, ECGFounder-based perceptual feature loss, Total Variation smoothing, and Pearson correlation loss to preserve ECG morphology and diagnostic signal structure.
- Built a patient-level experimental pipeline on 12 paired ear/chest ECG recordings with 36,407 two-second segments, preventing data leakage via subject-level train/validation/test splits and 90% overlap training windows.
- Achieved 0.8508 Pearson correlation on an unseen test patient and improved reconstruction to 0.9065 with decoder-only personalization, demonstrating subject-specific adaptation for wearable ECG monitoring.
- Profiled edge-deployment efficiency with 13,684 parameters and 13.44M MACs per 2-second window, motivating neuromorphic shift-add execution for ultra-low-power wearable AI systems.

Edge AI Product Recognition via 16-Stream RTSP (DeepStream & Jetson Nano) *Project Leader*

DOAN-DUC/DEEPSTREAM-YOLOV8-JETSON-NANO-16RTSP *Oct. 2025 - Dec. 2025*

- Built a high-throughput 16-stream real-time object detection pipeline on NVIDIA Jetson Nano using DeepStream, GStreamer, Docker, TensorRT, and YOLOv8n.
- Evaluated edge model optimization strategies, including INT8/FP16 quantization, knowledge distillation, and architecture fine-tuning to meet tight latency and memory constraints.
- Benchmarked real-time throughput across multi-camera RTSP configurations, ensuring robust detection accuracy at distances up to 10 meters.

Multi-tier Electronic Component Packaging AI Control System *Project Leader*

DOAN-DUC/OSCO-OBJECT-SCANNING-AND-CHECKLIST-OPTIMIZATION *Jan. 2026 - Feb. 2026*

- Developed a custom YOLOv8n model to identify and track 11 classes of electronic components in a complex 2-tier packaging pipeline.
- Engineered an 'Anchor-based Mapping' algorithm to synchronize and track component states in real-time across 4 concurrent camera streams.
- Designed and deployed automated QA logic to detect missing components, process deviations, and misplacements, triggering immediate alerts.

Skills

Programming	Python, C/C++
Deep Learning Frameworks	PyTorch, TensorFlow/Keras
ML / Data Libraries	Scikit-learn, NumPy, Pandas, Matplotlib, Seaborns
AI Research	Computer Vision, OpenCV, YOLO, CNNs, SNNs, KANs, RAG, Model Quantization, Knowledge Distillation
Systems & Deployment	Linux/Ubuntu, Docker, NVIDIA DeepStream, GStreamer, TensorRT, NVIDIA Jetson Nano
Tools	Git, Jupyter Notebook
Hardware	ESP32, Arduino
Languages	English (Intermediate), Japanese (Elementary)

Certificates

Mar. 2026	**Convolutional Neural Networks**	*DeepLearning.AI*
Sep. 2025	**Introduction to Deep Learning & Neural Networks with Keras**	*IBM*
May. 2025	**Machine Learning with Python**	*IBM*
Sep. 2025	**Getting Started with AI on Jetson Nano**	*NVIDIA*
Sep. 2025	**Getting Started with Machine Learning at the Edge on Arm**	*Arm Education*
Dec. 2025	**Gemini Certified University Student**	*Google*
Dec. 2025	**Global Consumer Intelligence Course 2025**	*Matsuo-Iwasawa Lab, UTokyo*
Aug. 2025	**Japanese-Language Proficiency Test (JLPT N4)**	*JEES / The Japan Foundation*
Sep. 2022	**IELTS Academic (Overall 7.0)**	*IDP / British Council*

References

Nguyen Huy Hoang, Ph.D *Hanoi University of Science & Technology*

SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING

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